Michael Tarlton

Physics-Informed AI Research · Neural Oscillations · Edge-Oriented Algorithm Design

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Profile

AI researcher and PhD candidate with a foundation in physics, mathematics and computational neuroscience. Develops neural oscillator-based learning models for temporal structure and periodicity detection, with an emphasis on algorithm design for computational efficiency in low-energy edge systems. Combines statistical modeling, scientific Python and controlled experimentation to investigate novel learning mechanisms in deep neural networks.

Core expertise

- **Physics and mathematical modeling:** statistical physics, complex dynamical systems, information theory, and Bayesian inference.
- **AI and algorithm design:** deep reinforcement learning, neuroscience-inspired neural design, temporal representations, periodicity discovery and hardware-oriented computational efficiency.
- **Experimental research:** simulation design, parameter sweeps, model evaluation, statistical analysis, numerical visualization and interpretation of model limitations.
- **Scientific computing:** Python, JAX, PyTorch, NumPy, SciPy, Polars, Matplotlib and MATLAB; research workflows using CI/CD pipelines and HPC environments.
- **Research and knowledge management:** project planning, oversight and execution using Agile methods; collaborative documentation using Notion, Obsidian and Markdown; experiment tracking with Aim and MLflow, and artifact and data management using Git LFS and SQL.

Research and professional experience

PhD Candidate — Artificial Intelligence

Oslo Metropolitan University · Applied Artificial Intelligence April 2022–present

- Research neuroscience-inspired learning algorithms based on oscillatory timing mechanisms, connecting neural dynamics with reinforcement learning and periodicity discovery.
- Develop Striatal Beat Frequency Automata: oscillator-based models using local weight updates to identify and learn temporal stimuli, with low-energy edge computation as an intended application.
- Evaluate trade-offs between learning performance and computational activity in hardware-oriented algorithms, focusing on local learning, sparse activation and simple update mechanisms.
- Published an AAMAS extended abstract and an IEEE/EMBS Neural Engineering conference paper as first author.
- Design and run controlled simulations to evaluate learning mechanisms, compare parameter settings and identify failure modes.
- Communicate research through technical writing and digital content; teach data visualization and supervise student projects.

Genetics Data Researcher

Norwegian Centre for Mental Disorders Research, University of Oslo January–April 2022

- Conducted statistical genomic research using genome-wide association study data in secure research-computing environments.

ML Data Scientist

Atlas Wearables (acquired by Peloton) December 2017–July 2018

- Designed experiments and analyzed biometric and movement data for proprietary machine-learning methods in the Motion Genome project.

Senior Support Technician

ServiceLedger · Lake Dallas, Texas March 2013–April 2016

- Provided desktop and network support in a Microsoft environment.
- Managed SQL databases and worked with XML stylesheets, report publishing and web applications.

Education and selected research

MSc in Neuroscience — 2021

Norwegian University of Science and Technology · Kavli Institute, Roudi Group

- Investigated network-topology inference using Ising models, Bayesian model selection and an information-based selection criterion.
- Simulated networks and compared inferred structures against ground truth across topology, parameter and sample-size conditions.
- Used MATLAB and NTNU's IDUN HPC environment for computational experiments linking statistical physics and neuroscience.

BSc in Physics · BA in Mathematics — 2017

University of North Texas

- Scientific computing, statistical astrophysics and information theory.
- Undergraduate research examined active-galactic-nucleus emission-line contributions to photometric bands using astronomical survey data.

Multidisciplinary astronomical-data research

xREZ Art + Science Lab · University of North Texas

- Contributed astronomical-data acquisition, analysis and scientific contextualization to *Instrument: One Antarctic Night*, a multidisciplinary visualization and sonification project.
- Prepared scientific data for an interactive environment using MySQL and SciPy; coauthored the resulting 2018 publication.

Selected publications and thesis

- **Tarlton et al. (2025).** "Investigation of Novel SBF-Automata Architecture for Periodicity Finding Solutions at Edge Systems." *IEEE/EMBS Conference on Neural Engineering*, pp. 1382–1387. https://doi.org/10.1109/NER61569.2025.11589137
- **Tarlton et al. (2024).** "Neurological Based Timing Mechanism for Reinforcement Learning." *AAMAS*, extended abstract, pp. 2504–2506. https://www.ifaamas.org/Proceedings/aamas2024/pdfs/p2504.pdf
- **Michael Tarlton (2021).** "Novel Model Selection Criterion for Inference of Ising Models." *Norges teknisk-naturvitenskapelige universitet.* Master's thesis. https://hdl.handle.net/11250/2783337
- **West et al. (2018), including Tarlton.** "Experiencing a Slice of the Sky: Immersive Rendering and Sonification of Antarctic Astronomy Data." *Electronic Imaging*. https://doi.org/10.2352/ISSN.2470-1173.2018.03.ERVR-449

Languages

English — native · Norwegian — B1